

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 04
Course Title : Operating System
Course Code : CSE 210
Experiment Name : Shell scripting using conditional statement, loops and file operations

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Question 1 – Take two numbers as input a & b.

If $b \leq 0$ print Error. Give positive number.

Else

$m = a/b$

$n = a \% b$

Print quotient and reminder

Steps to solve –

- Open “Git Bash”
- Go inside the Operating System folder
 - `cd “Operating System”`
- Create a shell script named “t.sh”
 - `touch t.sh`
- Open t.sh using command vi
 - `vi t.sh`
- Write the required shell script code and save the file
- Give execute permission to the file
 - `chmod u+x t.sh`
- Run the shell script file
 - `./t.sh`
- Enter the numbers, the conditions and arithmetic operation runs.
- Give the result as an output.

Code –

```
1  echo Enter first number:
2  read a
3  echo Enter second number:
4  read b
5  if [ $b -le 0 ] ; then
6  |   echo ERROR. Give positive number.
7  else m=$((a/b))
8  |   n=$((a%b))
9  |   echo Quotient: $m
10 |   echo    Reminder: $n
11  fi
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter first number:
45
Enter second number:
98
Quotient: 0
Reminder: 45
```

Question 2 – Take an integer number as input. If n is odd “1st Time”. If n is even & in the inclusive range of 2 to 5, print “Next Time”. If n is even & in inclusive of 6 to 20, print “First Time”. If greater than 20, print “Next Time 2”

Steps to solve –

- Open “Git Bash”
- Go inside the Operating System folder
 - cd “Operating System”
- Create a shell script named “t.sh”
 - touch t.sh
- Open t.sh using command vi
 - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
 - chmod u+x t.sh
- Run the shell script file
 - ./t.sh
- Enter a number, the conditions and arithmetic operation runs.
- Give the result as an output.

Code –

```
1 read -p "Enter a number " n
2 if [  $((n\%2)) -ne 0$  ] ; then
3     echo 1st Time
4 else
5     if [ $n -le 5 ] && [ $n -ge 2 ] ; then
6         echo Next Time
7     elif [ $n -le 20 ] && [ $n -ge 6 ] ; then
8         echo First Time
9     else echo Next Time2
10    fi
11 fi
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number5
1st Time
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number10
First Time
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number 10
First Time
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number 8
First Time
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number 55
1st Time
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
Enter a number 40
Next Time2
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$
```

Question 3 – Calculate using switch

Enter two numbers a & b. Press '+' to add two numbers, '-' to subtract, '*' to multiply and for division '/'.

Steps to solve –

- Open “Git Bash”
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 - cd “Operating System”
- Create a shell script named “t.sh”
 - touch t.sh
- Open t.sh using command vi
 - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
 - chmod u+x t.sh
- Run the shell script file
 - ./t.sh
- Enter two numbers a & b and t as operational character , the conditions and arithmetic operation runs.
- Give the result as an output.

Code –

```
Operating System > $ t.sh
1  read a t b
2  case $t in "+")
3      echo $((a+b));;
4      "-" )
5          echo $((a-b));;
6      "*" )
7          echo $((a*b));;
8      "/" )
9          echo $((a/b));;
10 esac
11
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
4 + 9
13
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
9 - 4
5
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
10 / 2
5
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
2 * 5
10
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$
```

Question 4 – Consider 1 to 20 (1 diff). If there's any number that is divided by both 3 and 5, break. Also when found 12, continue the process.

Steps to solve –

- Open “Git Bash”
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 - cd “Operating System”
- Create a shell script named “t.sh”
 - touch t.sh
- Open t.sh using command vi
 - vi t.sh
- Write the required shell script code and save the file
- Give execute permission to the file
 - chmod u+x t.sh
- Run the shell script file
 - ./t.sh
- Give the result as an output.

Code –

```
$ t.sh
Operating System > $ t.sh
1  for i in {1..20}
2  do
3      if [  $((i\%3)) -eq 0$  ] && [  $((i\%5)) -eq 0$  ] ; then
4          echo Numbers divided by 3 and 5: $i
5          break
6      fi
7      if [ $i -eq 12 ] ; then
8          echo Found 12
9          continue
10     fi
11     echo numbers are $i
12 done
13
```

Output –

```
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/Operating System$ ./t.sh
numbers are 1
numbers are 2
numbers are 3
numbers are 4
numbers are 5
numbers are 6
numbers are 7
numbers are 8
numbers are 9
numbers are 10
numbers are 11
Found 12
numbers are 13
numbers are 14
Numbers divided by 3 and 5: 15
```